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BIM and ontology-based knowledge management for dam safety monitoring

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ABSTRACT

Dam Safety Monitoring Systems (DSMSs) are crucial in evaluating the operational state of dams. However, heterogeneous monitoring data distributed from multiple sources during the dam operation lacks a unified data integration method and prohibits knowledge extraction and intelligent analysis, which currently poses a labor-intensive task. To address this issue, a solution relying on Building Information Modeling (BIM) and domain ontologies is proposed. Specifically, a DSMS domain ontology (OntoDSMS) is developed by comprehensively collecting domain knowledge and extracting context information from the dam information model. Furthermore, a rule-based reasoner and SPARQL queries are implemented. The proposed approach facilitates the effective integration of dam safety monitoring information while reducing retrieval time effectively compared with traditional databases. A case is illustrated to demonstrate the feasibility and practicality of the proposed approach.

1. Introduction

Water is exploited on a large scale as a clean, non-polluting, renewable resource. Therefore, numerous dams have been built to exploit the water resource fully and rationally, playing a vital role in constructing China's national economy. However, the complex working environment of a dam involves many hidden dangers that affect its safe operation. Indeed, these hidden dangers, if not detected in time, will significantly impact the safe operation of the dam, and once the dam breaks, it will cause irreversible harm to the social security and ecological environment. Therefore, it is mandatory to build a comprehensive multi-dimensional Dam Safety Monitoring System (DSMS) that controls and assesses the properties of the dam [1]. Various sensors have been deployed to detect hidden dangers, collecting real-time operational data [2], e.g., temperature and deformation. However, while a dam operates, the monitoring information collected by the sensors accumulates, significantly increasing the data management and analysis burden. Moreover, as the complexity of DSMSs increases, the data formats and heterogeneity are also increasing rapidly, inevitably raising the information sharing and integration complexity. Thus, a series of 'information islands' have emerged. Additionally, dynamic data generated in the ongoing DSMSs is fragmented and updated frequently, increasing the labor of manually analyzing the dynamic characteristics of the data. As a result, the existing DSMSs deal only passively with predefined static information and cannot deal with dynamic information intelligently, suffering from information integration and management deficiencies. Therefore, it is essential to improve the information management to sufficiently meet user's specific needs and facilitate the effective integration of dam safety monitoring information.

With the rapid advancement of informatization, Building Information Modeling (BIM) has been widely used in structural health monitoring and is dedicated to facilitating information exchange and interoperability [3]. BIM can describe the spatial relationship straightforwardly and intuitively, providing various effective solutions and practical guidance [4]. Indeed, current research has proved the benefit of integrating information in ongoing projects. For instance, Hou et al. [5] proposed a BIM-based visual warning management system for structural health monitoring integrated with a Long Short-Term Memory (LSTM) network. This system associated sensors with BIM elements to spatially visualize the sensors that corresponded with monitoring information and provided a basis for remote structural health monitoring and safety management. O'Shea et al. [6] designed a structural health

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monitoring system for a historic offshore lighthouse using BIM to visualize the output of the structural health monitoring integrating sensors. Valinejadshoubi et al. [7] developed a BIM-based data management system for the structural health monitoring of modular buildings. Their setup increased the efficiency of the structural condition assessment process to monitor the state of buildings in near real time.

However, current BIM implementations suffer from various drawbacks. First, the Industry Foundation Classes (IFC) has a flexible method of building information relying on an open neutral data format that effectively integrates heterogeneous data. However, complicated data schemas result in a lack of clear semantic mapping between entities and relationships. At the same time, multiple data structures may point to the same information, significantly limiting the automatic reasoning analysis extracted from the context information in a BIM environment [8]. Second, for most parts, BIM contains static data information lacking the capability to process dynamic data automatically. Finally, most BIM systems integrate information by linking IFC with a database, and this strategy is insufficient to improve information management.

Recently, combining BIM and ontology has gained wide attention, as it affords logical reasoning and standardized query function to address the issues mentioned above [9]. The ontology originated from the field of philosophy and gradually expanded to the field of information technology, involving syntax and semantic standards that humans and computers commonly understand. It is a collection of terms describing a domain with a hierarchical organizational structure and serves as the skeleton and foundation of the domain knowledge. The latter is described as triple sets in a domain ontology. The accurate description of knowledge based on the ontology prevents ineffective communication between humans and machines, ensuring that software works uniformly and in a predictable manner while maintaining a good collaboration with other software. Hence, ontology has significantly contributed to advancing information management capability by adding semantic information.

Web Ontology Language (OWL) [10] is a new ontology language developed by World Wide Web Consortium (W3C). Inheriting the Resource Description Framework (RDF), Resource Description Framework Schema (RDFS), and some early ontology languages, OWL assists machines in automatically processing and integrating information available on the Web [11]. The first specification proposal was recommended by W3C for OWL in 2004 and has been superseded by OWL 2, released in 2012. OWL is a declarative language rather than a programming language that describes information in a logical way that machines can understand. Therefore, machines can reason the knowledge expressed in OWL 2 ontology by verifying the knowledge consistency and explicitly expressing hidden knowledge [12].

To enhance the capability of traditional DSMSs for semantic description, logical reasoning, and data sharing, this paper establishes a DSMS domain ontology (OntoDSMS) based on the information-rich BIM. In addition to shortening the information retrieval time, the accuracy of the search results is improved. Hence, the intelligent and efficient integration of monitoring information can support the inference analysis of the operational status of a dam. The main contributions of the proposed scheme are as follow.

1) Extracting context information from a BIM environment and establishing a mapping between a Relational Database (RDB) and an ontology utilizing a virtual RDF graph.
2) Establishing a domain ontology, which stores multi-dimensional and fragmented data information as ‘one graph’. This strategy supports various management functions associating dynamic monitoring data with predefined static information in real time.
3) Establishing a rule-based reasoner and extending the function of SPARQL, which enhances the ability of logical reasoning and autonomic analysis. This combination infers valuable relationships and properties in big data. Simultaneously, this combination provides a consistent query result.

The remainder of the paper is organized as follows: Section 2 reviews the related BIM and ontology applications and identifies the differences between databases and ontologies. Section 3 introduces in detail the process of modeling OntoDSMS, while Section 4 develops a method for mapping a relational database and an ontology. Section 5 applies the OntoDSMS in a case study involving a concrete dam, and finally, Section 6 concludes this work and discusses the limitations and future research directions.

2. Literature review

Ontology, as the core of the Semantic Web, has contributed to the integration, extraction, and management of fragmented information. BIM-based ontology technology has been widely applied and studied in various construction projects to manage knowledge better. Indeed, domain ontology has been developed by many experts and scholars, with the corresponding studies mainly focusing on three aspects: data integration, knowledge management, and performance evaluation.

2.1. Application of ontology-based BIM

Building a domain ontology based on information-rich BIM enhances information sharing and reuse through structured information. Numerous studies on data standards and semantic selection have been conducted. Shen et al. [13] established a prefabricated building construction monitoring system based on ontology technology in a BIM environment. The ontology technology was applied in Revit software through plug-in integration to share, reuse, and accumulate safety risk management knowledge in the construction of prefabricated buildings. Pauwels et al. [14] obtained an OWL ontology for IFC (ifcOWL) from the EXPRESS schemas of IFC. An EXPRESS-to-OWL converter was constructed in their study. Farghaly et al. [15] constructed a safety ontology aligned with IFC and developed a semantic mapping approach between the developed ontology and ifcOWL. Using space ontology and layout transformation operations, Suter et al. [16] modeled multiple space views for schematic building design. This method semi-automatically converted the room-based building data created in the BIM authoring system into a multi-view spatial model. The advantage of ontology lies in providing a common framework for constructing data networks. Therefore, combining information-rich BIM and ontology can facilitate interoperability efficiently without misunderstanding and avoid data loss between different machines.

The implicit factual knowledge exists in a BIM environment objectively but cannot be directly extracted and utilized. This results in a low degree of information interconnection between the contexts. Based on the BIM environment, context information can be extracted to establish knowledge management ontology and compensate for the lack of semantic logical relationships of BIM. In [17–20], the class hierarchy in ontology defines the external framework of knowledge management, with the attributes determining the internal structure of knowledge management. Ait-Lamallam et al. [17] suggested an ontological approach to integrating operation and maintenance information in BIM for road infrastructure. It integrated operation and maintenance information within the IFC schema and ensured road infrastructure sustainability. Ren et al. [18] aligned BIM and ontology for information retrieval and reasoning in value for money assessments. This scheme was dedicated to retrieving information from a BIM environment with an ontology knowledge base to make the project and financial management methodology more compelling by facilitating decision-making. Martino et al. [19] developed a rule-based ontology system, which made inferences upon ontologies that originated from the IFC schema. This strategy identified real estate units and their use destination starting from BIM. Collinge et al. [20] presented a BIM-based construction safety risk library that facilitated tacit and explicit knowledge sharing in a BIM environment by linking security risks to processing through different risk scenarios.

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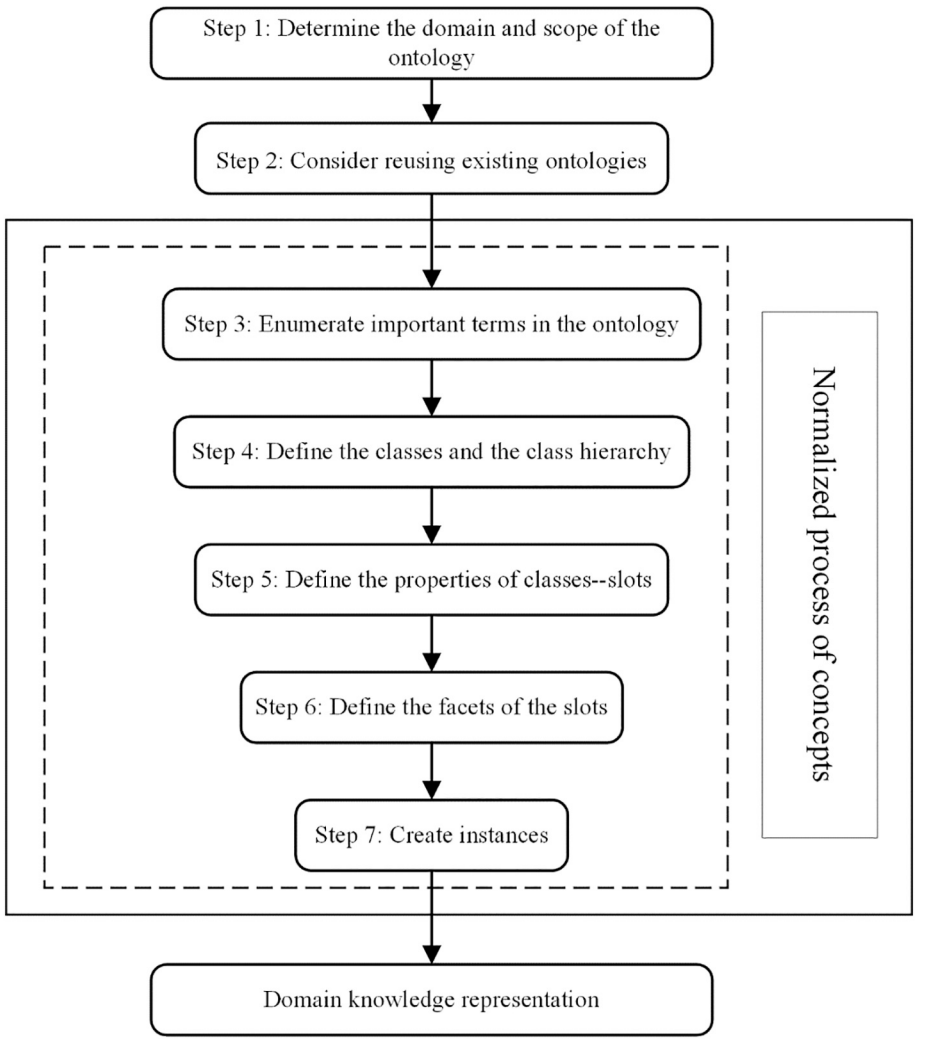


Fig. 1. Top-down modeling process using the Noy & McGuiness approach.

Establishing BIM application programs with inference support for performance analysis and calculations based on BIM data according to rules in regulations or standards has become a typical approach [21]. Jiang et al. [22] proposed a new approach that combined BIM and ontology, named green building evaluation ontology (GBEOntology), to facilitate intelligent green building evaluation. Li et al. [23] suggested a BIM-enabled semantic approach for automated safety checks in subway construction. It integrated heterogeneous data in a BIM environment and enabled automated safety checking through SPARQL-based reasoning. Jiang et al. [24] developed a gray-box checking technique and a BIM-based automated code compliance checking methodology that leverages ontology. This scheme assists code compliance by identifying substandard designs according to regulatory documents and improving design accuracy before construction. Zhang et al. [25] introduced a method for automatically checking the compliance of BIM-based building designs with relevant codes and regulations. This method relied on a transformer-based language model to learn semantic representations of the information in BIM and regulatory documents integrating automated IFC-regulatory information.

2.2. Ontology versus database

The database is the most widely used form of information storage, especially the RDB. However, an ontology has appeared as an alternative to a database [26]. The ontology has gradually developed from the field of philosophy to the field of information technology, which is considered a key element in enhancing domain-specific information. Compared with traditional databases, ontologies can improve information retrieval efficiency and search result accuracy. In practical engineering, databases and ontologies are complementary [27].

2.2.1. UNA, CWA, and OPA

Although databases and ontologies are employed to represent knowledge and information, these present some substantial differences.

First, a database adopts the Unique Name Assumption (UNA) [28], where the definition of the term asserts that different names correspond to different entities in the real world. On the contrary, an ontology recognizes multiple names for the same entity.

Second, two different assumptions exist in ontologies and databases besides UNA: Closed Word Assumption (CWA) and Open Word Assumption (OWA) [29]. CWA is used by systems that contain complete information, such as the RDB. According to CWA, there are three cases: 1) any unknown statement is False, 2) any object is True if it is declared True, and 3) any object is False if neither is declared True nor False. For example, if no relationships are declared between the entities in a relational database, the search result is bound to be an empty set. OWA is often applied to systems that contain incomplete information, such as the ontology. Unlike CWA, if there is no specific declaration that a relationship is True or False, then the relationship is not True nor False but unknown relying on OWA. This means that the relationship might be

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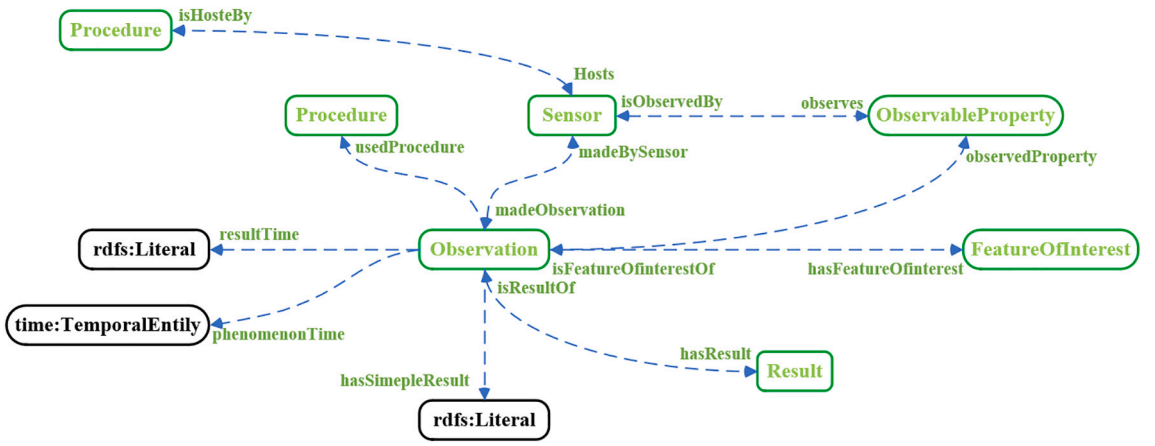


Fig. 2. Classes and relationships involved in Observation (SOSA).

True or False before more information is supplied, but no conclusion can be drawn until then [14]. Hence, hidden implicit information can be inferred based on rules in an ontology.

The difference between OWA and CWA determines the different attention of databases and ontologies. A database focuses on efficient information storage, while an ontology focuses on information semantics and understanding. In other words, a database is created as an effective data warehouse, whereas an ontology was formed for better communication and interoperability, and as a communication bridge between people and machines [27].

Reviewing the existing literature shows that although BIM and ontology have been applied in various construction projects, studies on managing and integrating the appropriate information for DSMSs are limited. This is because such implementations face some obstacles, the main ones being:

1) An extensible domain knowledge representation ontology applicable to a specific DSMS has not been defined. Due to the different requirements in various construction engineering tasks, it is difficult to unify the standards and information transmission for DSMSs.
2) The conversion method between the database and ontology is lacking. Nevertheless, it is worth creating a mapping between the existing database and the established domain ontology to reduce a considerable amount of repetitive work significantly.
3) Generic ontology cannot efficiently handle the relationship between dynamic data and the corresponding sensors. They cannot fully support essential project management functions [30].

This paper proposes a solution that exploits BIM and domain ontologies to address these issues. Specifically, an OntoDSMS is created by comprehensively collecting domain knowledge and extracting context information from BIM. Moreover, an effective data mapping between relational data and ontology is constructed, and a rule-based reasoner and SPARQL queries are implemented. The proposed approach facilitates the information management of DSMSs and makes information accessible, practical, and understandable.

3. Development of the OntoDSMS

The ontology model can be formally described as a set Onto = {URL, C, OP, DP, I}, where URL represents the unique identity of an ontology, C denotes a set of classes, OP is the object property that contains the relationship between two instances, DP is the data property that technically describes the details of a class or instance, and I represents a concrete instance of a series of classes [31]. Hence, it is necessary to define the implementation purpose and application scenarios first, and then the critical concepts and axioms in the domain should be described before developing a domain ontology.

The logical construction of a domain ontology mainly includes top-down and bottom-up approaches. The methods of constructing a domain ontology include TOVE [32], IDEF5 [33], Skeleton [34], KACTUS [35], SENSUS [36], METHONTOLOGY [37], and Noy & McGuiness approach [38]. The Noy & McGuiness approach developed by Stanford University School of Medicine, is considered to be relatively mature and is currently the only method that strictly conforms to the five basic principles of Gruber's ontology construction [39]. In the proposed method, the OntoDSMS adopts a top-down strategy through the Noy & McGuiness approach. Fig. 1 depicts the top-down modeling process using the Noy & McGuiness approach.

3.1. Step 1: Determine the domain and scope of the ontology

The first step of the Noy & McGuiness approach is defining the domain and scope of the ontology. It is an effective strategy to minimize the number of domain knowledge and concepts for the complex knowledge structure of dam safety monitoring. The proposed model aims to enhance the capability in information integration, knowledge sharing, and independent analysis, ensuring that the people involved can quickly find the relevant information. Therefore, the OntoDSMS contains complete objects, attributes, professional terms, and relationships in the traditional DSMS.

3.2. Step 2: Consider reusing existing ontologies

The next step is to extend and improve the existing resources for specific domains and tasks. Many ontologies in the form of documentation are imported into the development environment in use, allowing existing ontologies to be extended as the smallest units of modularity.

Semantic Sensor Network (SSN) is a standard ontology scheme recommended by W3C for describing sensors, observations, procedures, features of interest, samples, and actuators [40]. Following a horizontal and vertical modular architecture, SSN contains a lightweight but self-contained SOSA (Sensor, Observation, Sample, and Actuator). In the proposed method, some class hierarchies and axioms of the SSN and SOSA ontology are extended and modified. The classes and relationships related to Observation in the SOSA ontology are described in Fig. 2.

Typically, the form of the ontology representation is usually not important. In some existing semantic resources, knowledge serialization methods differ from OWL, which cannot be directly used in a specific domain. However, it is not different to transform the two serialization methods through knowledge extraction and fusion. For example, Pauwels et al. [9] integrated IFC and OWL architectures and constructed converters to generate ifcOWL ontologies from IFC. This method allows user technologies to represent structural data using a good IFC standard.

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Table 1
Classes established from the structural perspective.

Class	Description
LeftBankDam	The part of a dam which is on your left hand when you stand on the top of the dam facing the flow of the river
RightBankDam	The part of a dam which is on your right hand when you stand on the top of the dam facing the flow of a river
DrainGate	The part of a dam which is used to drain water
CrossSection	The sections in a dam
Element	The meta-unit of a component, e.g., columns, walls, dam sections
Material	The material type of a component, e.g., different grades of reinforced concrete

The conversion is approved by buildingSMART, and the latest ifcOWL version can be found on the buildingSMART international website [41].

3.3. Step 3: Enumerate important terms in the ontology

The third step is to enumerate essential terms in the ontology. The top-down development process starts with the most general concepts in the domain. To explain the domain knowledge clearly, it is necessary to list as many concepts as possible that the system wants to state or explain. The listed concepts and terms need to be declared or explained, regardless of whether the concepts overlap and how they are expressed, i.e., classes and attributes, as the essential thing is to obtain a comprehensive list of terms. For instance, the critical terms in DSMS include deformation monitoring, seepage monitoring, and environmental quantity monitoring. The measurement standards related to monitoring results include relative displacements, crack opening displacements, and uplift pressures. These terms related to structural physical attributes include elastic modulus, material information, and installation information.

3.4. Step 4: Define the classes and the class hierarchy

To date, the comprehensive terms related to the identified domain have already been listed, with some terms referring to the classes in ontology, while others describe the relationships between the classes. Hence, the classes and class hierarchy are defined in step 4. The approaches to developing a class hierarchy are as follows [34]: (1) The top-down approach that starts with the most general concepts defined and then specializes in explicit subclasses of the concepts. (2) The bottom-up approach, which starts with the most specific class defined and then extracts features to form the general concepts, and (3) the combination development process that combines top-down and bottom-up approaches to define salient concepts and then generalizes and specializes them appropriately.

Considering the DSMS complexity, the top-down approach is adopted to avoid duplicate work and knowledge inconsistencies. Considering the gravity dam, for example, the dam structure is relatively simple without complex component classification. According to the sensor distribution, the top-down dam structure is constructed. In the dam structure, the top-level parent class is named Structure, and the subclasses of Structure can be subdivided according to the scale of a specific dam, e.g., LeftBankSection, DrainGate, and RightBankDam. Then attribute classes like Material and Element are established from the perspective of component attributes. Structure can also be divided into fine-grained subclasses, whereas DrainGate can be divided into Pier, Gate, and other subclasses. However, the proposed method focuses on integrating multi-source heterogeneous data from dam safety monitoring systems. Thus, some irrelevant class topologies have been omitted. Table 1 lists the classes from the structural perspective.

Generally, the failure symptoms of dams can be summed up as abnormal deformation, abnormal seepage, severe changes of stress and strain within the structure, etc. Therefore, according to [42], the classes related to dam safety monitoring can be divided as follows:

Table 2
Classes established from the safety monitoring perspective.

Class	Description
EnvironmentQuantity	Monitoring the upstream water level, temperature, etc.
Transformation	Monitoring the size, distribution and change law of the deformation of a dam
Seepage	Monitoring the effect of seepage on the dam body and foundation
StressStrainAndTemperature	Monitoring internal stress and temperature changes
SensorType	Type of the sensor
System	A unit of abstraction for pieces of infrastructure
Platform	An entity that hosts other entities, particularly Sensors
Senor	A subclass of System, as well as the parent class of specific monitoring items
Observation	A collection of observations
ObservableProperty	A collection of observed properties. Including relative displacements, absolute displacements, cumulative displacements, water level, opening and closing displacements, seepage flow, strain, temperature, etc.
Abnormal	A collection of sensors whose working status is abnormal

Table 3
Properties and slots in an OntoDSMS.

Type	Property	Domain	Range
ObjectProperty	hasSection	LeftBankDam RightBankDam DrainGate	CrossSection
	isSectionOf	CrossSection	LeftBankDam RightBankDam DrainGate
	hasElement	LeftBankDam RightBankDam DrainGate	Element
	isElementOf	Element	LeftBankDam RightBankDam DrainGate
	hosts	LeftBankDam RightBankDam DrainGate	Sensor
	isHostedBy	Sensor	LeftBankDam RightBankDam DrainGate
	observes	Sensor	ObservableProperty
	isObservedBy	Observation	Sensor
	madeObservation	Sensor	Observation
	madeBySensor	Observation	Sensor
	hasType	Sensor	SensorType
	isTypeOf	SensorType	Sensor
	observedProperty	Observation	ObservableProperty
	hasMaterial	Element	Material
DatatypeProperty	hasResult	Observation	xsd:float
	reultTime	Observation	xsd:dataTime
	hasName	Sensor Element Material	xsd:string
	hasManufactures	Sensor Element Material	xsd:string
	hasElevation	Sensor	xsd:float

EnvironmentQuantity, Transformation, Seepage, StressStrainAndTemperature, etc. Considering the reusability of the existing ontologies in step 2, some core classes of SSN and SOSA are extended, while some axioms and attributes can be inherited from the framework of SSN and SOSA. The classes established from the safety monitoring perspective are listed in Table 2.

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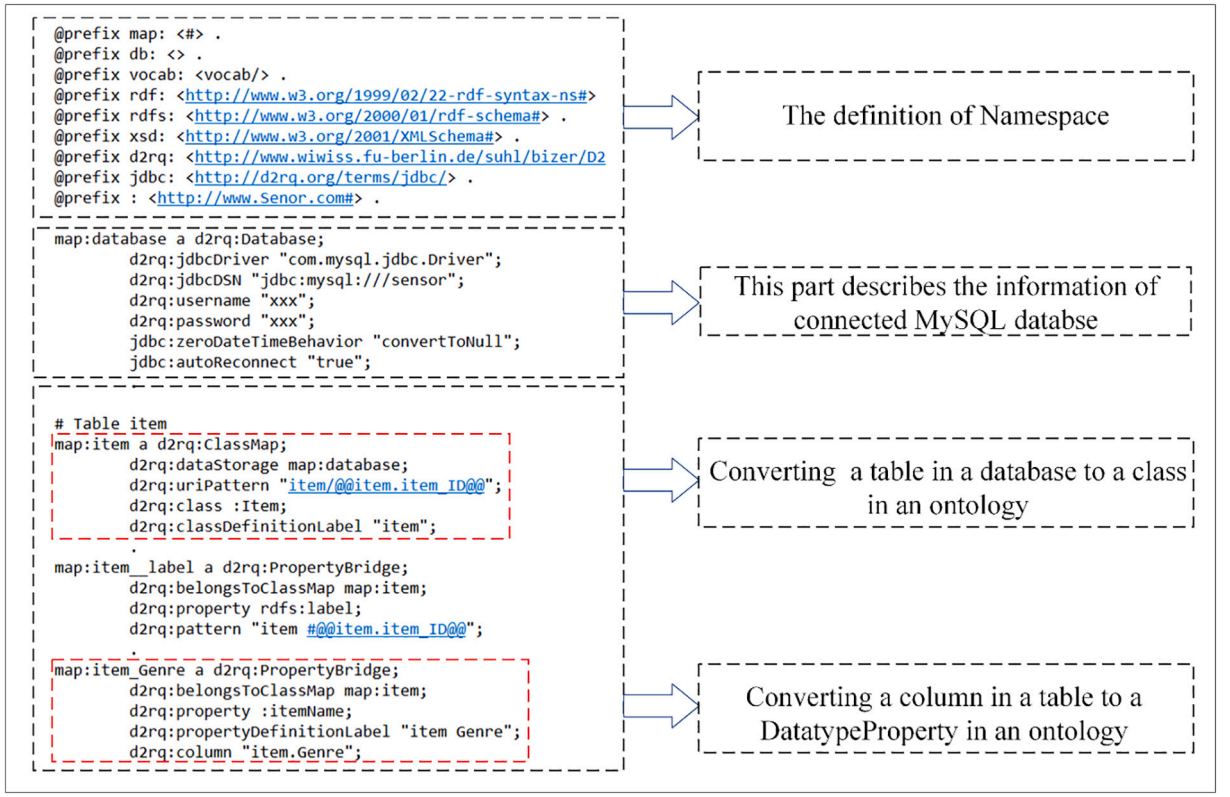


Fig. 3. Schema of the modified mapping file (partial view).

3.5. Step 5: Define the properties of classes-slots

After step 4, we have already selected some terms listed in step 3 as the classes. Most of the remaining terms are the properties of these classes. By now, the conceptual system with an external framework has been established. However, it does not reflect the internal relationships with adequate information. Therefore, the internal structure and class attributes need to be considered, with these attributes becoming slots attached to the classes. Slots are the descriptors for describing the properties of classes and instances [43]. The OWL properties include OWL: ObjectProperty and OWL: DatatypeProperty [44], where the former describes relationships between individuals and the latter describes the relationships between individuals and values. The structure and the facets of the slots are described in Table 3.

3.6. Step 6: Define the facets of the slots

A slot has different facets that describe the value type, allowed values, the number of the values (cardinality), and the other feature values the slot can take. A slot typically contains the value type, domain, and range. The cardinality of the slot defines the number of the values, which allow specification of a minimum and maximum cardinality, or any number that accurately describes number of slot values. For example, if the Material slot describes the specific component composition, it can be defined as the type of concrete in the component with different proportions. The slot value types include String, Number, Boolean, and Enumerated. For example, the classes related to safety monitoring have slot-hasName, and the value is AP1. The value type of hasName is String. The allowed classes for slots of type Instance are often called a range of a slot. The classes to which a slot is attached or the classes which property a slot describes, are called the domain of the slot [38].

3.7. Step 7: Create instances

The last step is to fill the individual instances into the completed framework by (1) selecting a class, (2) creating a single instance of that class, and (3) filling the slots. The single instance involves a concrete executor instantiated by the framework.

4. Converting the information stored in a RDB to RDF

The buildings, sensors, and personnel topologies in a DSMS are complex and heterogeneous. Moreover, data information is usually stored locally, which makes it challenging to share and dynamically modify between different applications, prohibiting a conducive data exchange and interoperability. A RDB has a sufficient capacity as a container storing data but fails to extract the relationships effectively. Keyword-based search is a common and typical search method in a RDB. For example, when complex many-to-many relationships exist in the system, the retrieval efficiency in a RDB is weaker than in an ontology [45], as the results may exclude some documents of interest when the keywords are too specific. Conversely, search performance may degrade when the keywords are too general, and the results may contain too many irrelevant documents [46]. Hence, converting an existing relational database into a corresponding ontology improves system management and eliminates repetitive work.

Mapping the information representations in a RDB to information representations in an ontology is trivial. This work implements the transformation through the open-source platform named D2RQ [47]. A virtual RDF graph accessing to the relational database is one of the essential features of D2RQ, offering RDF-based access to the content of the RDB without replicating it into an RDF store. According to D2RQ [48], the following operations can be implemented.

1) a non-RDF database can be queried using SPARQL.

2) the content of the database can be accessed as Linked Data over the Web.

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Table 4
Main information of dam safety monitoring sensors.

Monitoring items		Number of sensors
Deformation	Horizontal displacement	12
	Vertical displacement	11
	Crack	22
Seepage	Uplift pressure	24
	Foundation seepage	10
	Seepage around the dam	20
	Leakage	15
Stress, strain, and temperature	Stress	25
	Strain	25
	Temperature	12
Environment quantity	Upstream water level	1
	Downstream water level	1
	Rainfall	1

3) custom dumps of the database can be created in RDF formats for loading into an RDF store.

4) the Apache Jena API can access information in a non-RDF database.

D2RQ supports Oracle, MySQL, PostgreSQL, SQL Server, and HSQLDB databases. Different methods have a consistent mechanism that generates a mapping file through mapping language and constructs a virtual data structure as a virtual RDF graph. Moreover, D2RQ translates the RDF query operations into SQL statements and implements the operations in a RDB. The D2RQ mapping files are RDF documents written in Turtle syntax. The basic principles of mapping are as follows.

1) The tables in a RDB correspond to the classes in an ontology.
2) The columns of the table correspond to the DatatypeProperties in the ontology.
3) The rows of the table correspond to the Instances in the ontology.
4) The values of the table cells correspond to the Literals in the ontology.

A predefined mapping file can be generated automatically according to the databases in the D2RQ platform. Modifying the file is necessary to keep the glossary consistent with the established ontology and map the data to the ontology. The mapping file is written in the mapping language, while the modified local schema of a mapping file is illustrated in Fig. 3.

5. Case study

The case study involves a concrete dam located in China. In this case, the dam is divided into 12 sections and contains 179 sensors. Table 4 reports the deployment of the monitoring instruments. The framework of managing monitoring information based on an OntoDSMS is depicted in Fig. 4. To combine BIM and ontology, the links between the objectively existing information model are established and the conceptual rules for extracting knowledge from big data are established. This process is shown as Ontology Establishment and Data Integration in Fig. 4. The SPARQL queries and inference functions are extended to extract valuable relationships hidden in isolated data and provide consistent results. This process is shown as Apache Jena-based Engine and Application in Fig. 4.

5.1. Framework of the OntoDSMS

Considering the DSMS complexity, the top-down logic and the Noy & McGuiness approach are used to define the framework of the OntoDSMS. To accelerate the process of developing the ontology, the open-source ontology editor protégé is employed to establish the domain ontology.

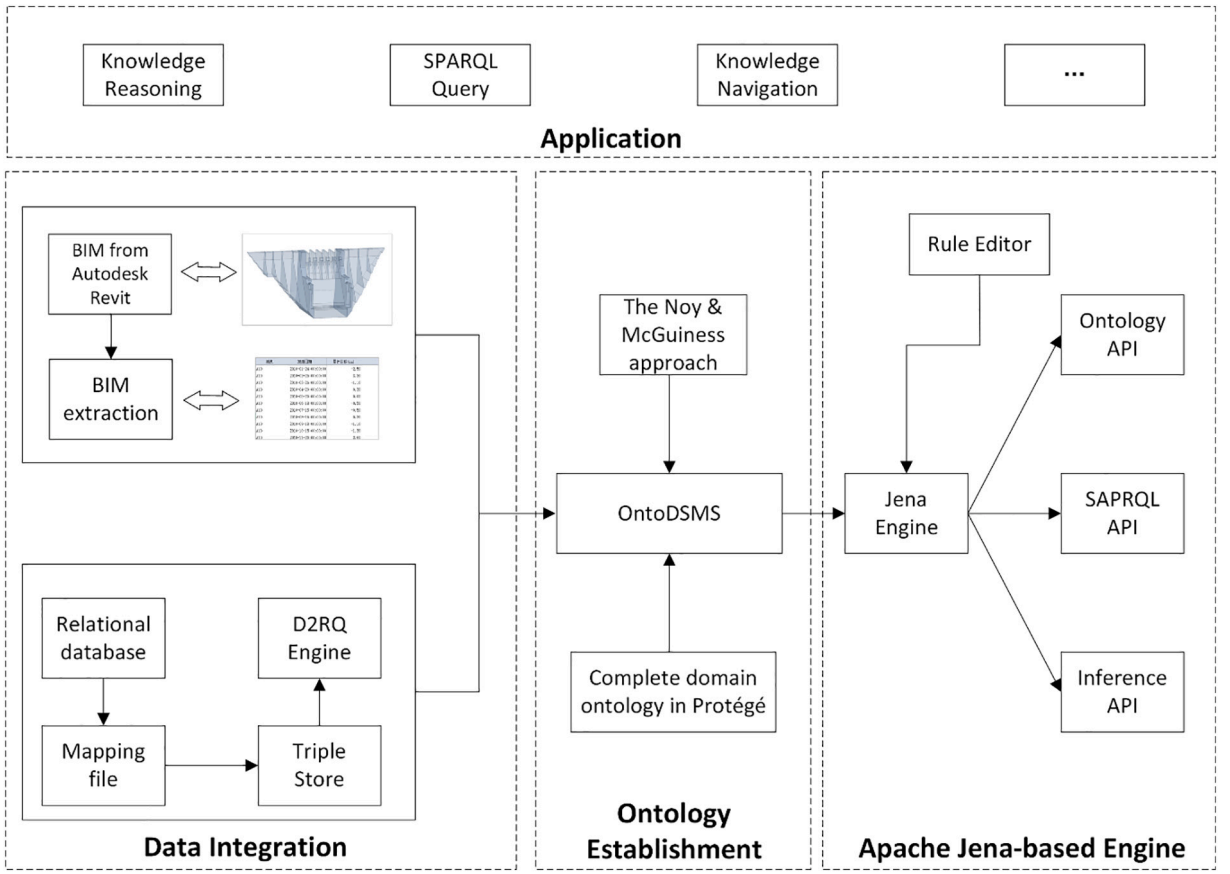


Fig. 4. Framework of managing monitoring information based on the OntoDSMS.

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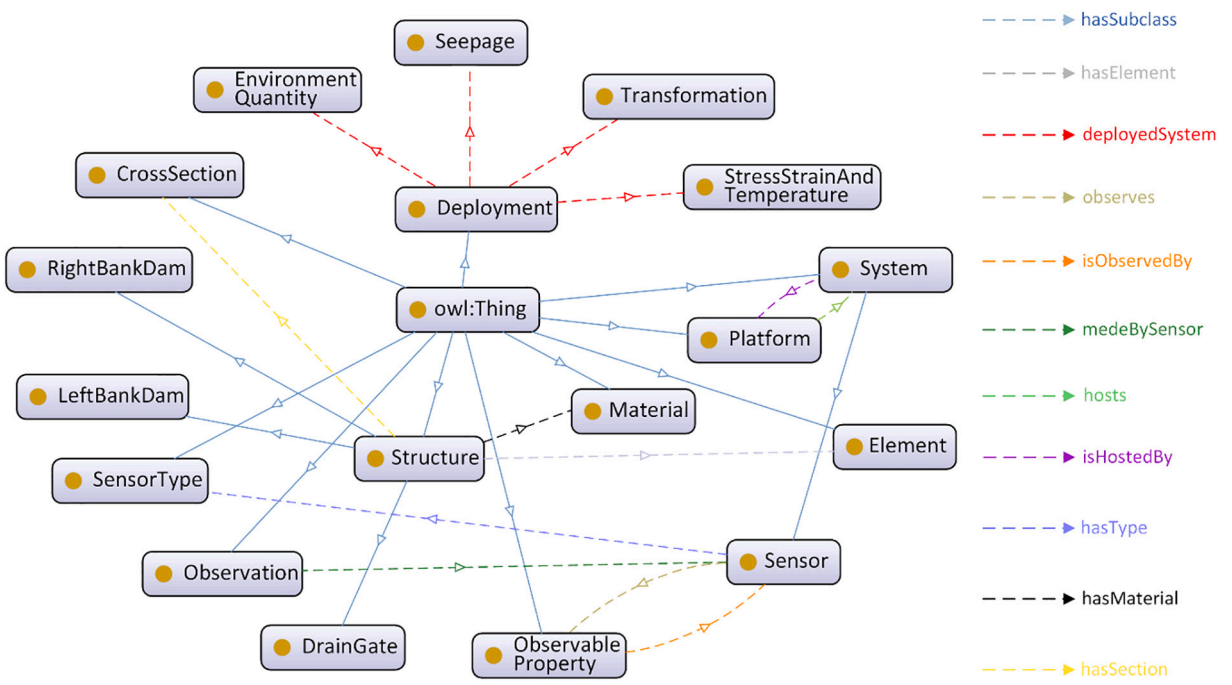


Fig. 5. Class hierarchy of OntoDSMS established in Protégé.

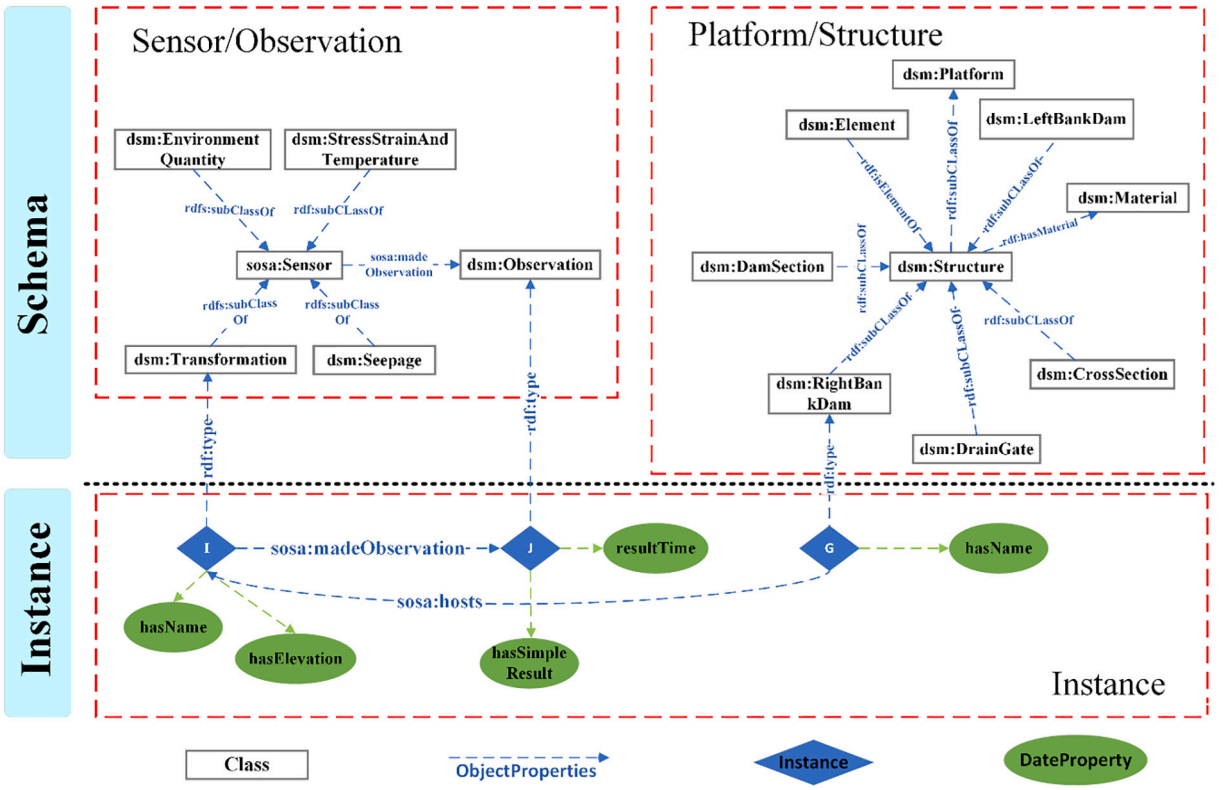


Fig. 6. Knowledge graph related to Sensor, Observation and Structure.

The friendly GUI operation of Protégé provides many knowledge model architectures and actions, which can be used to create, visualize, and operate various forms of ontologies, supporting users to create a knowledge model and fill data flexibly. In addition, it is highly extensible and can test a SPARQL query ahead of time by using some embedded tools to implement rule-based reasoning and ability. Fig. 5 illustrates the local class hierarchy of the developed OntoDSMS, while Fig. 6 depicts the corresponding relationship between the schema and instance of the OntoDSMS. Considering the reuse of existing ontologies, the core classes of SSN and SOSA were extended, and some axioms and attributes were inherited from the framework of SSN and SOSA.

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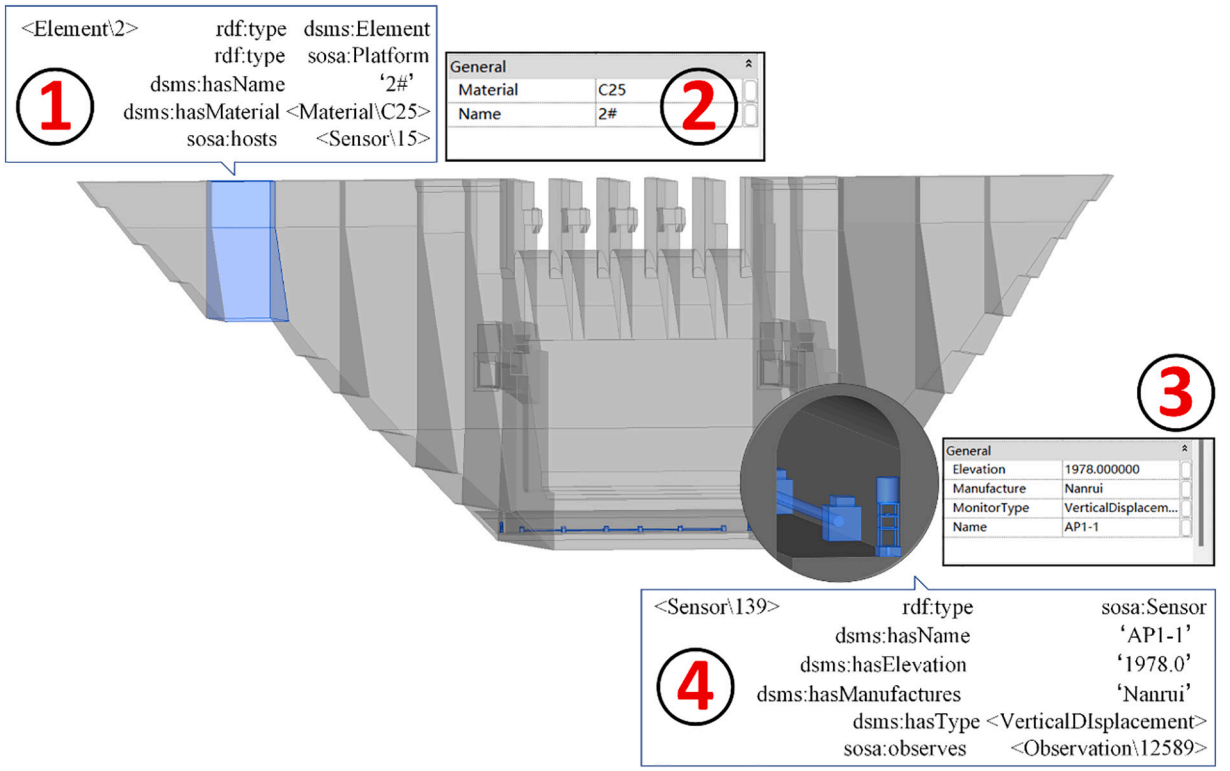


Fig. 7. Extracting context information from the BIM environment and the corresponding description in OWL.

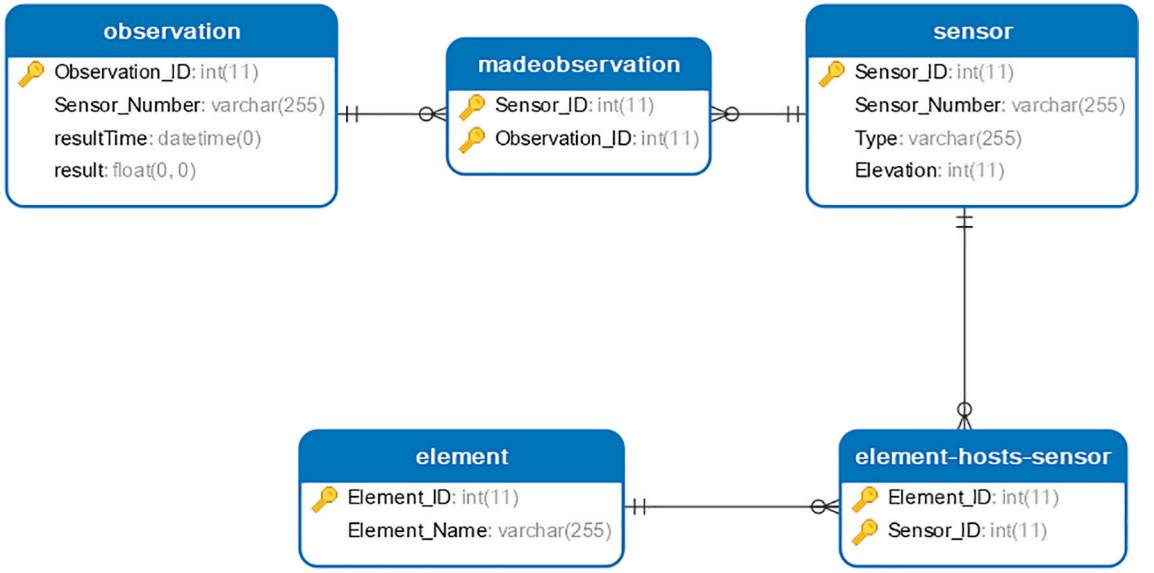


Fig. 8. Table structures in MySQL (partial view).

5.2. *Information extraction from BIM*

This section aims to extract context information from the BIM environment and convert it to the RDF format. Fig. 7 illustrates the information-rich program built in Autodesk Revit, one of the most widely used BIM core modeling software, which is adopted to provide a basic BIM environment. The advantage of Autodesk Revit is creating parametric components through a graphical family editor. Revit assembles complete projects from positioned elevations and axis networks. A complete project stores rich parameterized information in the form of graphics. Therefore, predefined context information is linked to graphics by creating different parameters in the modeling stage. From the Structure perspective, some parameters like Name and Material are set. From the Sensor perspective, some parameters like Elevation, Manufacture, MonitorType, and Name are set. As presented in parts 2 and 3 in Fig. 7, the related contextual information can be triggered in Revit when clicking on the model. The corresponding descriptions of that information in the ontology are presented in parts 1 and 4. In addition, all context information can be exported into a spreadsheet in a structured format and stored in a MySQL database. Section 4 presents the mapping between the database and the ontology.

In addition to the context information extracted from the BIM

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SPARQL Query 1

```
SELECT ?S ?N ?O ?T
WHERE {
  ?E a :Element.
  ?E dsm:hasName '4#'.
  ?E sosa:hosts ?S.
  ?S dsm:hasName ?N.
  ?S dsm:hasType 'Displacement_Vertical'.
  ?S sosa:madeObservation ?O.
  ?O sosa:hasResult ?R.
  ?O sosa:resultTime ?T.
  FILTER(?T >  "2015-02-28T00:00:00Z"^^xsd:dateTime)
}
```

SPARQL Query 2

```
SELECT (COUNT (?S) AS ?Number)
WHERE {
  ?E a :Element.
  {?E :hasName '4#'}
UNION
  {?E :hasName '5#'}
UNION
  {?E :hasName '6#'}
  ?E :hosts ?S.
}
```

Fig. 9. SPARQL query 1 and SPARQL query 2.

environment, the MySQL database is the other primary distributed data source. A mapping between MySQL and ontology is required to integrate the historical data, real-time collected data, and contextual information. Some of the table structures in the MySQL database are illustrated in Fig. 8.

5.3. *Consistency queries extending SPARQL function*

To publish ontology data for searching and sharing on the network directly, we construct consistent queries, applications of semantic web and ontology data based on Apache Jena, an open-source semantic web framework widely used and documented for the semantic web framework. Jena provides RDFS, OWL, general-purpose rule inference machines, and SPARQL endpoint and can be run as a web application or an embedded server. To illustrate the effectiveness of OntoDSMS as a semantic middleware, four representative query examples are provided, with the query results proving the flexibility and ability to retrieve specific information.

The first example shows how to query the monitoring data of the vertical displacement sensor deployed in the dam section No.4# after 2015-2-28. As shown in Fig. 9, Query 1 is executed by the code, with the detailed query code grammar presented in [49]. The dam comprises 12 sections, with each installing some sensors to monitor the operation state. Traditional DSMSs contain sensor installation information without considering the relationships between the sensors and the components. The relationships must be obtained by comparing the installation documents of the sensors and components or manually querying the drawings. This query lists a series of variables representing the case of interest as?S,?N,?O, and?T. The detailed structural information in the system allows users to query in terms of global views.

The second example shows how to query the number and type of sensors installed in dam sections No.4#, No.5#, and No.6#. The dam structure is complex, and even professionals may need to check the installation documents or drawings to obtain the installation information. To deal with the complex deployment and help ease the communication for non-professionals, the installation information can be obtained rapidly in Query 2, as shown in Fig. 9.

The third example shows the query to access the sensors in a crack monitoring system whose working state is abnormal, including the time of occurrence and the component being observed. Traditional RDB cannot realize the query without using an external program. The iterative calculation results are time-consuming for the enormous historical monitoring data involving hundreds of thousands of entries. In this case, the relationship between sosa: Sensor and dsm: Abnormal is not explicitly defined, and the generic reasoner rules-based is created to infer which sensors are the subclass of dsm: Abnormal. In the related rules, the observed allowed maximum and minimum values are given as warning indicators, with the inference engine triggered when the result is larger or less than the indicators. The corresponding code and rules are presented in Fig. 10, and the formulation of the indicators can be found in [42].

The fourth example reveals how to extend and modify data when the monitoring information is updated. The OntoDSMS is a multifaceted and dynamic system that emphasizes dynamic updates, real-time interactions, and timely responses. Therefore, the established ontology must be extended and modified as new data is generated. To address this

SPARQL Query 3

```
SELECT ?N1 ?S ?N2 ?O ?T
WHERE {
  ?E a :Element.
  ?E dsm:hasName ?N1
  ?E sosa:hosts ?S.
  ?S a dsm:Abnormal .
  ?S dsm:hasName ?N2.
  ?S sosa:madeObservation ?O.
  ?O sosa:hasResult ?R.
  ?O sosa:resultTime ?T
}
```

Rule

```
[[ruleMaximum: (?x dsm:hasType 'Crack')
               (?x dsm:hasMaxIndicator ?y)
               (?x sosa:madeObservation greaterThan(?z,?y))
               -> (?x a dsm:Abnormal)]]
[[ruleMinimum: (?x dsm:hasType 'Crack')
               (?x dsm:hasMinIndicator ?y)
               (?x sosa:madeObservation lessThan(?z,?y))
               -> (?x a dsm:Abnormal)]]
```

Fig. 10. SPARQL query 3 and rule involved.

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SPARQL Query 4

```
INSERT DATA
{
<http://www.sensor.com#Sensor/180> a :Senor;
                dsm:hasName 'Ap1-1a';
                dsm:hasElevation '1978.0';
                dsm:hasManufactures 'Nanrui';
                dsm:hasType 'VerticalDisplacement'
}
```

SPARQL Query 5

```
WITH <http://www.sensor.com>
DELETE {?S dsm:hasName 'B2-1'}
INSERT {?S dsm:hasName 'BP2-1'}
WHERE{
        ?S dsm:hasName 'B2-1'
}
```

Fig. 11. SPARQL query 4 and SPARQL query 5.

Table 5
Comparison results of SPARQL and SQL queries.

	Result count of SPARQL	Result count of SQL	Response time of SPARQL (s)	Response time of SQL (s)	Number of entities
Query 1	2989	2989	0.26	2.72	492,875
Query 2	39	39	0.19	1.57	
Query 3	2	2	2.35	7.26	

issue, an actuator based on SPARQL Update language is set up, triggered to update the ontology when the data is updated. The SPARQL Update is an updated language for RDF graphs derived from the SPARQL Query language. The Update operations are performed on a collection of graphs in a Graph Store, and the operations update, create, and remove RDF graphs in a Graph Store [50]. As illustrated in Fig. 11, a new sensor is added to the system containing its number, location, type, and monitoring information by executing the Query 4 code. The number of the sensor numbered B2–1 is modified to BP2–1 by executing the Query 5 code. More rules can be found in [50].

To compare the relational database and ontology, corresponding SQL-based queries were provided to verify the results. Table 5 reports the number of entities in the above results and the retrieval time. The number of entities included in both systems is 492,875, while all other data in the table are retrieved 10 times under the same conditions, and the average value is calculated.

6. Conclusions

The operation data of a dam directly presents the operational characteristics and the evidence to judge whether a problem occurs. Therefore, it is crucial to establish a feedback mechanism through DSMSs for a timely and intuitive understanding of the operational status of the dam. This paper develops a BIM-based domain ontology for better managing knowledge. Specifically, we construct a knowledge network comprising triples that provide the semantic relationships between the sensors, sensors and structures, and structures, solving the information island problem. Extracting context information from the BIM environment and mapping it using a virtual RDF graph reduces the ontology modeling time and improves the working efficiency. Indeed, we demonstrate that the enhanced function of SPARQL is more effective compared with traditional methods, affording autonomous reasoning from multi-source heterogeneous data to discover hidden relationships or new knowledge.

The innovations of this paper are as follows. First, a DSMS domain ontology is proposed to integrate domain knowledge and construct semantic-BIM systematically. Our method compensates for the lack of semantic logical relationships in BIM. Second, SSN and SOSA are applied to a traditional DSMS, and the multi-dimensional and fragmented information is stored as ‘one graph’. This solution enhanced the integration and interoperability of the traditional DSMS. Third, we develop a data mapping method between RDB and RDF, significantly reducing repetitive work and improving management efficiency. Finally, expanding the SPARQL queries improves the logical reasoning ability of the traditional DSMS. Moreover, we provide several typical cases to demonstrate the ability and precision of the proposed method.

However, this work has limitations. First, only structural non-geometric attributes are considered during modeling BIM-based ontology, which may impose an increased burden on the system when the graphical data are updated. Second, it is still a challenge to realize the dam structure damage identification, anomaly monitoring, and safety evaluation of monitoring data based on ontology, although the current knowledge management of DSMSs has overcome the interoperability problem. These deficiencies shall guide future research work.

CRediT authorship contribution statement

Yuhang Zhou: Investigation, Data curation, Formal analysis, Writing - original draft. **Tengfei Bao:** Investigation, Resources, Supervision, Project administration. **Xiaosong Shu:** Writing – review & editing. **Yueyang Li:** Software, Visualization, Methodology, Validation. **Yangtao Li:** Software, Visualization, Methodology, Validation.

Declaration of Competing Interest

The authors declare no conflict of interest.

Data availability

No data was used for the research described in the article.

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